

COMPUTER NETWORKS CHEAT SHEET

BASICS

Topology: Layout of network (star, bus, ring).

LAN/MAN/WAN: Small, city, large/global networks.

Point-to-Point vs Broadcast: One-to-one vs one-to-all.

DATA COMMUNICATION

Components: Sender, Receiver, Message, Medium, Protocol.

Modes: Simplex, Half-duplex, Full-duplex.

Media: Guided (wired), Wireless (air).

SWITCHING

Circuit: Fixed path, reserved.

Packet: Data split, dynamic path.

Message: Full message, slow.

MULTIPLEXING

Multiplexing: Many signals one channel.

FDM/TDM/WDM: Frequency, Time, Light.

Modem: Digital ↔ Analog.

ARCHITECTURE

Protocol: Rules of communication.

Service: Layer functionality.

OSI: 7 layers.

TCP/IP: 4 layers.

TCP/UDP/IP

TCP: Reliable, slow.

UDP: Fast, no guarantee.

IP: Addressing + routing.

DATA LINK

Framing: Divide into frames.

Error Detection/Correction: CRC, Hamming.

ACK: Confirmation.

Sliding Window: Multiple frames.

Token Bus/Ring: Controlled access.

CONTROL

Flow Control: Sender vs receiver speed.

Multiple Access: Shared medium.

CONGESTION

Congestion Control: Avoid overload.

Traffic Shaping: Control rate.

Choke Packet: Reduce speed.

Load Shedding: Drop packets.

ROUTING

Routing: Path finding.

Deterministic/Adaptive: Fixed/dynamic.

Centralized/Distributed: One vs many routers.

ALGORITHMS

Distance Vector: Neighbor sharing.

Link State: Full network view.

Path Vector: Full path.

PROTOCOLS

RIP: Simple.

OSPF: Fast.

BGP: Internet.

MULTICAST

Multicast: One to many.

Address: Group IP.

Routing: Efficient delivery.

Protocols: DVMRP, MOSPF, PIM.